1. Clinical Question & Core Hypothesis

Clinical Question

CEACAM5 is upregulated in tumors and correlates with higher immune cell infiltration and MSI-H, yet high immune infiltration does not always translate into effective anti-tumor immunity or therapy response. 2) Built on the single provided path linking CEACAM5 → higher immune infiltration → MSI-H because it directly connects a tumor-expressed adhesion molecule to an inflamed, potentially immunogenic genomic background. 3) Supporting evidence (Tier 2 equivalent): MSI-H tumors are known to be immunogenic (high neoantigen load) and yet clinical responses vary, implying tumor-intrinsic modulators (like surface adhesion molecules) could alter immune cell localization/function. The ultrasound→gene expression edge was treated as a negative control (no causal effect). 4) I chose therapy_response as the endpoint because MSI-H and immune infiltration are most clinically relevant to immunotherapy responsiveness, making this the most problem-focused outcome. 5) Deliberately excluded additional genes or speculative immune checkpoints to keep the mechanistic question sharp; ultrasound and claims that ultrasound alters gene expression were excluded because the path states no evidence of causation?

Working Hypothesis

CEACAM5 upregulation in tumor cells promotes an immune-inflamed microenvironment characterized by high Tumor-Infiltrating Lymphocyte (TIL) density, but creates either retention of T cells at tumor cell surfaces or induction of T cell exhaustion/suppressive signaling, resulting in dysfunctional anti-tumor immunity. Consequently, tumors with high CEACAM5 expression and high TIL density will exhibit reduced objective therapy_response to immunotherapy compared with CEACAM5-low tumors with similar TIL densities. This effect is particularly pronounced in MSI-H tumors, where high neoantigen load coexists with CEACAM5-mediated modulation of immune Engagement. Predicted testable features include: (1) CEACAM5-high tumors show high TIL density but increased exhaustion markers (e.g., PD-1, LAG-3) and altered spatial distribution (peritumoral clustering with limited intratumoral cytotoxic synapses); (2) CEACAM5 expression correlates with poorer therapy_response to checkpoint inhibitors after adjusting for TIL density, tumor mutational burden, and PD-L1; (3) functional assays where CEACAM5 manipulation alters T cell activation in co-culture and CEACAM5 overexpression impairs cytotoxic killing in vivo under immunotherapy pressure.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	CEACAM5
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CEACAM5: CEACAM5 is upregulated in tumors and correlates with higher immune cell infiltration and MSI-H, yet high immune infiltration does not always translate into effective anti-tumor immunity or therapy response.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	MSI-H	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- MSI-H: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.